Hints: Combinatorics – 1. Counting and bijections

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Identify the three colours with $0, 1, 2 \in \mathbb{Z}/3$. For three colours a, b, c, when is "two equal, one different" true, in terms of $a + b + c \bmod 3$?

Hint 2. All equal or all distinct is exactly $a + b + c \equiv 0$; so the condition is that every face sum is non-zero mod 3. Consider the linear map $\mathbb{F}_3^{30} \to \mathbb{F}_3^{20}$ sending an edge colouring to its 20 face sums; the answer is 2^{20} times the size of a fibre, provided the map is surjective.

Hint 3. The map is the unsigned vertex–edge incidence map of the dual graph (the dodecahedron: faces are vertices, edges join adjacent faces). For a connected non-bipartite graph this map is onto over a field of characteristic $\neq 2$ (adjust values along a spanning tree, and use an odd cycle to fix the last degree of freedom). The kernel has dimension 10, so the answer is $2^{20}3^{10}$.

Problem 2

Hint 1. Note that the weights $1, 1, 2, \ldots, 63$ add up to 2017. Let N_c be the number of tuples with weighted sum $\equiv c$. Multiplying all x_i by a non-zero residue shows N_c is the same for all $c \neq 0$, and $N_0 + 2016N_1 = 2017!/1953!$. One more relation is needed.

Hint 2. Compute $S = \sum \omega^{x_0 + x_1 + 2x_2 + \cdots + 63x_{63}}$ over tuples of distinct elements, with ω a primitive 2017th root of unity; then $S = N_0 - N_1$. Remove the distinctness condition by Möbius inversion over set partitions of the 64 indices: a partition π contributes $\mu(\pi) \prod_{B \in \pi} \sum_x \omega^{w(B)x}$, where $w(B)$ is the total weight of block B.

Hint 3. An inner sum is 2017 if $w(B) \equiv 0$ and 0 otherwise, and only the whole set has weight divisible by 2017. So only the one-block partition survives, with $\mu = (-1)^{63}63!$, giving $N_0 - N_1 = -2017 \cdot 63!$. Solving: $N_0 = \frac{2016!}{1953!} - 2016 \cdot 63!$.

Problem 3

Hint 1. Write $N = \frac{10^a - 1}{9} \cdot 10^b$ with $a \geqslant 1$ ones and $b \geqslant 0$ zeros, $a + b \leqslant 2020$. Factor $2020 = 4 \cdot 5 \cdot 101$.

Hint 2. $4 \cdot 5 \mid N$ iff $b \geqslant 2$, and $101 \mid N$ iff $10^a \equiv 1 \pmod{101}$ iff $4 \mid a$ (the order of 10 mod 101 is 4). Count the pairs $a = 4k$, $2 \leqslant b \leqslant 2020 - 4k$: $\sum_{k=1}^{504}(2019 - 4k) = 508536$.

Problem 4

Hint 1. Use the Chinese remainder theorem: $x^2 - x = x(x - 1)$, and x, $x - 1$ are coprime.

Hint 2. $10^{2006} \mid x(x - 1)$ iff $x \equiv 0$ or $1 \bmod 2^{2006}$ and $x \equiv 0$ or $1 \bmod 5^{2006}$: four residues mod 10^{2006}, one of which is 0. The answer is 3.

Problem 5

Hint 1. Factor over $\mathbb{C}$: $(P + iQ)(P - iQ) = X^{2n} + 1$. What are the degrees and leading coefficients of the two factors, given $\deg Q < \deg P$?

Hint 2. $\deg P = n$, and $P + iQ$ has real leading coefficient c with $c^2 = 1$. Its roots are n of the $2n$ roots of $X^{2n} = -1$, and $P - iQ$ (its conjugate) takes the conjugate roots, so $P + iQ$ contains exactly one root from each conjugate pair. Conversely each choice gives valid real P, Q with $\deg Q < n$. The answer is $2 \cdot 2^n = 2^{n+1}$.

Problem 6

Hint 1. Try small grids ($2 \times n$, 3×3). Each coin moves along its diagonal; describe which sets of coins have moved (each coin moves at most once, since afterwards some square it needs is occupied).

Hint 2. Show by induction on moves that in every reachable configuration the moved coins form an "up-right closed" set: there is a monotone lattice path from the bottom-left to the top-right corner of the $(m-1) \times (n-1)$ board separating moved coins from unmoved ones, and conversely every such path is reachable. The answer is $\binom{m+n-2}{m-1}$.

Problem 7

Hint 1. With lengths $a_1 < a_2 < \ldots$, the endpoint is $x = a_1 - a_3 + a_5 - \ldots$, $y = a_2 - a_4 + \ldots$. For an endpoint in the closed first quadrant, determine which $n \pmod 4$ are possible by looking at the sign of the last horizontal and vertical pairs.

Hint 2. Only $n = 2$ and $n \equiv 2 \pmod 4$, $n \geqslant 6$, give first-quadrant endpoints. With $n = 2$ you get exactly the points with $0 < x < y$. With $n \geqslant 6$ you get $x = a_1 + (a_5 - a_3) + \ldots \geqslant 3$ and $y = a_2 + (a_6 - a_4) + \ldots \geqslant 4$, and $n = 6$ already realises every such point.

Hint 3. So the non-endpoints are: $x = 0$ (2012 points), $(1, y)$ with $y \leqslant 1$, $(2, y)$ with $y \leqslant 2$, and (x, y) with $x \geqslant 3$, $y \leqslant 3$. Total $2012 + 2 + 3 + 4 \cdot 2009 = 10053$.

Problem 8

Hint 1. Only residues mod 3 matter. Among $1, \ldots, 3k+1$ there are $k+1$ numbers $\equiv 1$, k numbers $\equiv 2$, k numbers $\equiv 0$. Zeros do not change the running sum mod 3 (but cannot come first).

Hint 2. Ignoring zeros, the running sum must avoid 0 mod 3, which forces the residue sequence $1, 1, 2, 1, 2, \ldots, 1, 2$ (or starting $2, 2, 1, \ldots$, impossible with these counts). Compute the probability that the non-zero residues come in this order and that the first number is not $\equiv 0$; the answer is $\frac{k!\,(k+1)!}{(3k+1)\,(2k)!}$.

Problem 9

Hint 1. Inclusion–exclusion with $A_i =$ maps $[n] \to [k]$ missing the value i; $|\bigcap_{i \in I} A_i| = (k - |I|)^n$.

Hint 2. A surjection is the same as a partition of $[n]$ into k non-empty blocks together with a bijection from blocks to $[k]$; hence the count is $k!\,S(n,k)$ and $S(n,k) = \frac{1}{k!} \sum_j (-1)^j \binom{k}{j} (k-j)^n$.

Problem 10

Hint 1. $f(n)$ counts tuples $(c_1, \ldots, c_n)$ of non-negative integers with $\sum c_k\, k! = n!$. Once $c_2, \ldots, c_n$ are chosen, c_1 is determined (if non-negative). Both bounds come from estimating $\prod_{k=1}^{n-1} \frac{n!}{k!}$.

Hint 2. Upper bound: $c_k \leqslant n!/k!$, so $f(n) \leqslant \prod_{k=1}^{n} (n!/k! + 1) \leqslant 2^n \prod_{k=1}^{n} n!/k!$. Lower bound: choose $c_k \in [0, \frac{n!}{n \cdot k!}]$ freely for $2 \leqslant k \leqslant n-1$ and $c_n = 0$; then the sum is at most $n!$ and c_1 completes it. This gives $f(n) \geqslant \prod_{k=2}^{n-1} \frac{n!}{n\,k!}$.

Hint 3. Use Stirling in the form $\log k! = k \log k - k + O(\log k)$: $\log \prod_{k \leqslant n} \frac{n!}{k!} = n \log n! - \sum_{k \leqslant n} \log k! = \frac{n^2}{2} \log n - \frac{n^2}{4} + O(n \log n)$. All the losses ($2^n$, n^n, the $O(n \log n)$ error) are of the form $n^{O(n)}$, which is what the constant C absorbs.